

Professional experience

MINES Paristech - PSL - CEMEF

SOPHIA-ANTIPOLIS, FRANCE

PhD Researcher

Jan '24 - Now

- Conducting doctoral research as part of the ERC-funded project CURE on novel coupling of computational mechanics and machine learning methods for hemodynamics prediction in intracranial aneurysms
- Developing several methods in Pytorch specific for Navier-Stokes equations: GNN, Deep Reinforcement Learning, Multigrid approaches
- Collaborating with an interdisciplinary team and physicians at the Hospital CHU Nice using real patient specific cases

Flaneer, a startup providing computer in the cloud for the creative industry

PARIS, FRANCE

Founder - Co-CEO & CTO

Sep '20 - Jan '24

- Co-founded and led a startup that provides cloud computing solutions for the creative industry, raising \$1.4M in funding from top global VCs and Angels. Recruited and managed a team of 8.
- Developed a full cloud Virtual Desktop Infrastructure (VDI) system with AI cost optimization, winning 5 RD awards from different government and private organizations serving over 10k+ users

CNRS, fluid dynamics team

PARIS, FRANCE

Applied Scientist

May '20 - Aug '20

- Conducted research on the applications of Deep Reinforcement Learning (DRL) for fluid mechanics and mesh optimization
- Created a framework that coupled DRL and fluid mechanics using Keras, Gym, and Fenics

Amazon, search relevance team

PALO ALTO, US

Applied Scientist Intern

Aug '19 - Feb '20

- Worked on the search team that helped customers find the best products among over 1 billion items on amazon.com
- Created and used advanced neural networks and transformer models that refined the relevance and ranking of search results. Achieved a 7.19% increase in NDCG@10 using listwise loss, transformers and textual features

Mines Paristech, fluid dynamics team

PARIS - NICE, FRANCE

Fluid mechanics and Reinforcement Learning Researcher

May '19 - Aug '19

- Development of DRL to several fluid mechanics problems. (Keras, Gym, and Fenics)
- Development of an open source project for Fluid Mechanics and DRL, [a blog post](#) and [a paper](#) with over 100s citations.

Mines Paristech, missions for H4D and IOMA

PARIS, FRANCE

Machine Learning Consultant

Jun '19 - Aug '19

- Case study and proof of concepts for these two companies, in order to launch two PhD's.
- Development of several models related to time series, data anonymization, NLP and computer vision. (Keras, Tensorflow)

Ministry of Culture, application referencing every cultural event in Paris

PARIS, FRANCE

Co-founder of Arts Around Me

Sep '17 - Mar '19

- Created an app in November 2017 about arts in Paris, and started to work with the Ministry of Culture in January 2018.
 - Used Machine learning and statistics to improve the Pass Culture recommendation algorithm.
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Education

MINES Paristech - PSL - CEMEF

PARIS & SOPHIA ANTIPOLIS, FRANCE

PhD student, Machine Learning for Fluid Dynamics

2024 - 2026

MINES Paristech - PSL

PARIS, FRANCE

Master Degree in Science & Executive Engineering

2017 - 2021

Lycee Sainte-Genevieve

VERSAILLES, FRANCE

Classe preparatoire MP*/MPSI - Mathematics and Computer Science Options

2015 - 2017

Peer-Reviewed Publications

A graph based machine learning method for fast and accurate simulations of intracranial aneurysms, under review at Nature

TRELLIS-Enhanced Surface Features for Comprehensive Intracranial Aneurysm Analysis, under review at Neuroscience Informatics

Automated discovery of finite volume schemes using Graph Neural Networks, under review at Nature Computational Science, by P. Garnier, J. Viquerat, E. Hachem

Training Transformers for Mesh-Based Simulations, under review at NeurIPS, by P. Garnier, V. Lannelongue, J. Viquerat, E. Hachem

MeshMask: Physics-Based Simulations with Masked Graph Neural Networks, published at ICLR 2025, by P. Garnier, V. Lannelongue, J. Viquerat, E. Hachem

Multi-Grid Graph Neural Networks with Self-Attention for Computational Mechanics, published in Physics of Fluids, by P. Garnier, J. Viquerat, E. Hachem

Evaluating Soccer Player: from Live Camera to Deep Reinforcement Learning, published in ICJAI by P. Garnier, T. Gregoir

A review on Deep Reinforcement Learning for Fluid Mechanics, published in Computers & Fluids, by P. Garnier, J. Viquerat, J. Rabault, A. Larcher, A. Kuhnle, E. Hachem

Projects & Teaching Experience

Fall 2024 & 2025: Introduction to Graph Neural Networks Graduate Student Instructor

Since 2018: Codingame - AI Contests: Development of artificial intelligence using machine learning, MCTS, MinMax and evolutionary algorithm for games. **Top 1% over 20 contests.**

2018: Creation of a robot to automatically harvest cauliflower: Created our own dataset after several fields visits and trained a ML model. Used a Raspberry PI to add the model to an existing robot.

Technical Skills & Hobbies

Computer Science: Python, Java, AWS, GCP, Pytorch, CI/CD. Led teams on both research project & software engineering products. Top 0.2% on **Codingame**.

Languages: French (*mother tongue*), English *fluent* (TOEFL 111/120), Mandarin Chinese (HSK 2).

Sports: Ultra-trail runner, 7+ sport climber, regatta in France & Hong-Kong

Mountaineering: Multi-pitch climber, TD Alpine itinerary, Ice climbing